

A Hierarchical ReAct Multi-Agent Pipeline for Residential Appliance Load Scheduling Under Dynamic Electricity Pricing

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Abstract

Residential households on dynamic electricity contracts can reduce costs by shifting flexible appliance loads to off-peak windows, yet coordinating multiple appliances across a volatile price curve requires automation that current single-agent LLM-based solutions fail to provide reliably. Existing implementations suffer from multi-appliance coordination failures, one-dimensional optimization signals, and evaluations limited to single-day scenarios.

This thesis designs and evaluates a hierarchical ReAct multi-agent pipeline built in LangGraph, comprising an Orchestrator, an Appliance Profiler, a Price-Aware Scheduler, and a Conflict Resolver. The study investigates to what extent the pipeline reduces scheduling cost compared to a single-agent baseline, which price-volatility conditions yield the greatest improvement, and which appliance characteristics drive cost savings. Evaluation uses REFIT Smart Home consumption profiles and Finnish Nord Pool FI spot prices across 48 scenarios, with performance measured as optimality gap relative to a MILP ground truth and assessed using the Wilcoxon signed-rank test. The methodology follows Design Science Research.

Keywords

multi-agent system, ReAct, LangGraph, home energy management, appliance scheduling, dynamic electricity pricing, Design Science Research

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Foreword

Work in progress.

Abbreviations

API	Application Programming Interface
DSR	Design Science Research
HEMS	Home Energy Management System
LLM	Large Language Model
MAS	Multi-Agent System
MCP	Model Context Protocol
MILP	Mixed Integer Linear Programming
Nord Pool FI	Nord Pool Finland Bidding Zone
ReAct	Reasoning and Acting
REFIT	Renewable Energy in the Future of an Innovative Technology
SLM	Small Language Model
SLR	Systematic Literature Review

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1. Introduction

Driven by dynamic pricing and smart grid infrastructure, an architectural shift is transforming how residential electricity consumption is monitored, scheduled, and optimized.The proliferation of smart metering infrastructure, the growing adoption of dynamic electricity pricing contracts, and the expanding penetration of renewable energy sources are together creating both an opportunity and a challenge for residential energy management. Households with dynamic spot-price contracts can substantially reduce their electricity bills by shifting flexible appliance loads to cheap off-peak windows. However, the cognitive burden of coordinating multiple appliance cycles across a volatile 24-hour price curve exceeds what most households can practically manage without automated assistance.

In Finland, this challenge has reached particular salience. Since October 2025, Finnish electricity consumers on spot-price contracts face 15-minute pricing intervals tied to the Nordic spot market (Nord Pool FI bidding zone), with intraday price ratios of up to 2.5 times between the cheapest and most expensive hours (Haakana et al., 2025). At the same time, research in LLM-based Home Energy Management Systems (HEMS) has advanced rapidly. Recent hierarchical multi-agent implementations demonstrate that language model reasoning can be applied to appliance scheduling tasks that previously required domain-expert rule encoding or computationally expensive optimization solvers (El Makroum et al., 2026). However, these implementations carry three documented limitations that this thesis directly addresses.

1.1 Background and motivation

The research journey toward optimizing residential energy consumption is shifting from rigid heuristic algorithms to autonomous, reasoning-capable AI architectures. Conventional HEMS solutions have historically relied on mathematical optimization solvers, rule-based schedulers, or classical machine learning models trained on historical consumption patterns (Beaudin & Zareipour, 2015). While effective in constrained and well-defined settings, these approaches require significant domain expertise to configure and do not generalize to novel price regimes.

The emergence of Large Language Models (LLMs) as general-purpose reasoning engines has opened new possibilities for HEMS design. Yao et al. (2023) demonstrate that LLMs equipped with the ReAct paradigm — a Thought-Action-Observation loop that interleaves reasoning with tool calls — can solve multi-step planning problems requiring dynamic information retrieval and constraint satisfaction. Applied to residential scheduling, a ReAct agent can reason over a price forecast, classify appliances by shiftability, identify the cheapest available time window, and verify constraint compliance through a traceable reasoning chain.

A secondary motivation concerns sustainable AI deployment. Small language models (SLMs) in the 7 billion parameter range consume 10 to 30 times less energy per inference than large frontier models while maintaining competitive task quality (NVIDIA, 2025). Since the Low Carbon Alert trainee project context prioritizes energy efficiency in the AI systems used to schedule household loads, a secondary evaluation axis compares the commercial cloud model GPT-4o-mini with gpt-oss-20b, an open-weight mixture-of-experts model (about 21 billion parameters, of which about 3.6 billion are active per token) run locally on a laptop, contributing to the research agenda on sustainable agentic AI deployment.

1.2 Research problem

Existing LLM-based HEMS implementations carry three documented limitations. Multi-appliance coordination fails in some model configurations due to context window overload. Optimization signals are one-dimensional, based solely on wholesale price without considering simultaneous load peak constraints. And evaluations are confined to single-day scenarios that leave robustness across price-volatility regimes untested (Jonah et al., 2026).

To address these limitations, the study is guided by one main research question and two sub-questions:

> **RQ:** To what extent does a hierarchical ReAct multi-agent pipeline reduce residential appliance scheduling cost compared to a single-agent baseline, and under which price-volatility conditions is the improvement most pronounced?
>
> **SQ1:** How should the pipeline be designed to handle multiple simultaneous appliance scheduling requests without redundant reasoning across agents?
>
> **SQ2:** Which appliance types benefit most from multi-agent scheduling, and what appliance characteristics drive the largest cost reduction?

The main RQ is deliberately non-binary. It asks how much and under which conditions, producing a richer and more nuanced answer than a binary yes/no finding. Together the three questions cover performance magnitude (RQ), design rationale (SQ1), and domain insight (SQ2).

1.3 Design Science Research approach

This study follows the Design Science Research (DSR) methodology as defined by Peffers et al. (2007). DSR is appropriate because the central activity is the iterative construction and empirical validation of a software artifact — the hierarchical ReAct multi-agent pipeline — rather than the observation of a pre-existing phenomenon. As Hevner et al. (2004) explain, DSR produces knowledge and understanding of a design problem and its

solution through the building and evaluation of an artifact. The artifact is the primary research contribution. A Systematic Literature Review (SLR) is conducted as DSR Phase 1 to confirm the research gap and ground all architectural decisions in prior work.

1.4 Structure of the thesis

This thesis is organized into seven chapters. Chapter 2 presents the literature review covering HEMS concepts, LLM-based agents for energy management, the ReAct framework, multi-agent LLM systems, current agentic AI trends, and the Finnish energy market context. Chapter 3 describes the Design Science Research methodology and evaluation strategy. Chapter 4 presents the system design and implementation. Chapter 5 presents evaluation results across 48 scenarios. Chapter 6 discusses findings, limitations, and future work. Chapter 7 concludes the thesis.

2. Literature Review

This chapter reviews the theoretical background and prior empirical work relevant to the design and evaluation of a hierarchical ReAct multi-agent pipeline for residential appliance scheduling. The review is organized into seven thematic areas. The chapter concludes by synthesizing the literature into a confirmed research gap that motivates the three research questions of this thesis.

2.1 Home energy management systems

A Home Energy Management System (HEMS) is a software or hardware system that monitors, controls, and optimizes residential electricity consumption to reduce cost, support grid stability, and improve energy efficiency (Beaudin & Zareipour, 2015). The core scheduling problem is the assignment of flexible appliance loads — those whose start times can be shifted without violating user preferences or physical constraints — to time windows where electricity prices are lowest, while ensuring that total household load at any moment does not exceed safe operational limits.

The literature classifies HEMS scheduling approaches into three broad categories. Rule-based systems apply expert-defined heuristics to trigger or defer appliance cycles based on price thresholds. Mathematical optimization methods including Mixed Integer Linear Programming (MILP) and dynamic programming compute provably optimal schedules subject to precise constraint definitions, but require all constraints to be expressed in mathematical form and cannot handle natural language user preferences (Moradzadeh & Tomsovic, 2013). Machine learning approaches learn scheduling policies from historical data but do not generalize well to novel price regimes (Mocanu et al., 2018). A key limitation shared across all three approaches is their inability to adapt dynamically to new constraint types or explain scheduling decisions in terms that non-expert household users can understand and trust.

2.2 LLM-based agents for energy management

The application of LLMs to energy management tasks has emerged as an active research area since 2023. El Makroum et al. (2026) present a hierarchical agentic AI architecture comprising one orchestrator agent and three specialist agents evaluated across three model variants: LLaMa 3.3 70B, Qwen 3 32B, and GPT-OSS 120B. Their system demonstrates cost reductions of 15–30% compared to unscheduled baselines on representative household scenarios. However, the system does not include thermal loads such as heat pumps, electric heating systems, or hot water storage, and evaluation is confined to a limited set of scenarios that do not systematically vary price-volatility regime.

Jonah et al. (2026) implement a single-agent ReAct HEMS that grounds one tool-calling agent in live half-hourly spot prices, weather forecasts, and household usage data. Their evaluation against a MILP ground truth reveals three explicit limitations: multi-appliance coordination failures in some model configurations, one-dimensional optimization signals based solely on wholesale price, and evaluation confined to single-day scenarios that leaves robustness across price-volatility regimes untested. These three limitations constitute the primary research gap this thesis addresses.

2.3 The ReAct framework

The ReAct (Reasoning and Acting) framework, introduced by Yao et al. (2023), proposes that LLM agents should interleave reasoning traces with action execution rather than separating reasoning and acting into sequential phases. In the ReAct paradigm, each agent step produces three outputs: a Thought — explicit reasoning about the current state and next action; an Action — a structured call to an external tool or API; and an Observation — the result returned by that tool. This loop repeats until the agent reaches a final answer.

Yao et al. (2023) evaluate ReAct against chain-of-thought reasoning and direct action baselines across four benchmark tasks. ReAct outperforms both baselines on all four tasks, with particularly large gains on tasks requiring dynamic information retrieval and multi-step planning — precisely the characteristics that distinguish HEMS scheduling from simpler single-step optimization tasks. The interpretability of the ReAct trace is a distinctive advantage for energy management applications: because each agent step records the reasoning behind every tool call and schedule decision, the full trace provides an auditable explanation of why a particular appliance was assigned to a particular time window.

2.4 Multi-agent LLM systems

Multi-agent LLM systems have been proposed as a solution to the context window and coordination limitations of single-agent designs. Han et al. (2024) survey the challenges and open problems in LLM multi-agent systems, identifying task allocation, memory management, and orchestration pattern design as primary research challenges. The survey confirms that role-specialized multi-agent designs — where each agent is responsible for one well-defined subtask and has access only to the tools needed for that subtask — consistently outperform generalist single-agent designs on complex multi-step tasks.

Hong et al. (2023) demonstrate role-based multi-agent orchestration in a software development context through MetaGPT, where agents act as product manager, architect, engineer, and QA reviewer. The MetaGPT architecture shows that structuring agent interactions through a shared state object eliminates the redundant context that causes coordination failures in single-agent designs — a finding directly relevant to the multi-appliance coordination failures documented in single-agent HEMS implementations.

2.5 Agentic AI trends in 2025–2026

Several 2025–2026 trends in agentic AI are directly applicable to this thesis. The most significant is the shift from single-agent to hierarchical multi-agent orchestration. Anthropic (2026) states that organizations in 2026 will be able to harness multiple agents acting together to handle task complexity that was difficult to imagine just a year ago. This shift is driven by the fundamental limitation that complex tasks exceed what any single agent context window can handle effectively.

The Model Context Protocol (MCP), introduced by Anthropic in November 2024 and donated to the Linux Foundation's Agentic AI Foundation in December 2025, has emerged as the industry standard for agent-to-tool communication (Anthropic, 2024). MCP provides a standardized, model-agnostic interface that enables AI agents to connect to external tools and data sources, eliminating the brittle custom integration code that previously characterized agent tool implementations. The LangChain MCP adapters library makes MCP tools directly compatible with LangGraph agents, enabling the price API and data reader in this thesis to be implemented as reusable MCP servers.

Human-in-the-Loop (HITL) design has been identified as one of the seven canonical agentic AI design patterns alongside ReAct, Reflection, Tool Use, Planning, Multi-Agent Collaboration, and Sequential Workflows (acecloud.ai, 2026). In this thesis, a lightweight HITL checkpoint where the Orchestrator presents the proposed schedule to the user for approval before the Conflict Resolver finalizes it is incorporated as an optional LangGraph interrupt node, aligning the design with current best practice for high-stakes autonomous decisions.

2.6 Finnish energy market context

Finland provides a directly relevant operational context for this research. The country operates within the Nordic spot electricity market (Nord Pool) and as of October 2025 has adopted 15-minute pricing intervals, making it one of the most granularly priced residential electricity markets in Europe. Approximately 31% of Finnish household electricity consumers currently hold spot-price-based electricity contracts that expose them directly to hourly and quarter-hourly price fluctuations (Haakana et al., 2025).

Intraday price volatility in the Finnish market is substantial. Analysis of Nord Pool FI historical data reveals that the cheapest hours typically sit around €0.033 to €0.046 per kWh, while morning demand peaks push prices to approximately €0.0815 per kWh — a 2.5 times ratio under normal conditions. During extreme events such as the 2022–2023 energy crisis, Finnish households experienced price spikes up to eight times higher than normal overnight rates (Numminen et al., 2026). These structural features make Finnish residential demand flexibility not merely a consumer cost issue but a national energy system challenge.

2.7 Literature synthesis and research gap

The literature reviewed across the preceding sections establishes three converging findings. First, HEMS scheduling is a multi-step, multi-constraint optimization problem that LLM-based agents can address more flexibly than rule-based or classical optimization approaches. Second, single-agent ReAct HEMS implementations carry documented, explicitly stated limitations in multi-appliance coordination, optimization signal dimensionality, and evaluation scope (Jonah et al., 2026). Third, multi-agent role-specialized designs consistently outperform single-agent baselines on complex multi-step tasks in other domains (Hong et al., 2023; Han et al., 2024), but no published system applies this design pattern to HEMS with a controlled comparison across multiple price-volatility regimes.

This gap is the direct motivation for this thesis. The study makes three contributions: a hierarchical ReAct multi-agent pipeline for residential appliance scheduling implemented in LangGraph; a quantified comparison across 48 evaluation scenarios spanning six price-volatility regimes with statistical significance testing and per-appliance breakdown; and a secondary comparison of the commercial GPT-4o-mini with the locally run open-weight gpt-oss-20b on scheduling quality, reliability and inference cost.

3. Research Methodology

This chapter describes the research methodology and explains how Design Science Research is applied in this thesis. It presents the rationale for the selected approach, the phase mapping to thesis activities, and the evaluation strategy including the statistical methods used.

3.1 Design Science Research methodology

This thesis follows the Design Science Research (DSR) methodology as defined by Peffers et al. (2007). DSR is a research paradigm focused on the design and evaluation of innovative artifacts that solve complex practical problems. As Hevner et al. (2004) argue, design science creates and evaluates IT artifacts intended to solve identified organizational problems, producing knowledge through the construction and evaluation of artifacts rather than through the observation of natural phenomena. DSR is the appropriate methodology for this thesis because the central activity is building and evaluating the multi-agent pipeline artifact, not observing a pre-existing system.

Artifacts in DSR are classified into four types: constructs, models, methods, and instantiations (March & Smith, 1995). In this thesis, the artifact is an instantiation — a working software implementation of a hierarchical multi-agent scheduling pipeline — supported by architectural models (the LangGraph state graph design) and constructs (the agent role taxonomy: Profiler, Scheduler, Resolver, Orchestrator).

Table 3.1 maps the six DSR phases defined by Peffers et al. (2007) to the specific activities and outputs in this thesis.

3.2 Evaluation strategy

The evaluation follows a simulation-based controlled design. No real sensors, buildings, or hardware are required. All inputs are drawn from publicly available datasets and APIs, and all evaluation scenarios are constructed from historical real-world data rather than synthetic simulations. The evaluation comparison is paired: the same 48 scenarios are run through the multi-agent pipeline, the single-agent baseline, and the MILP solver.

The primary evaluation metric is the schedule optimality gap, defined as:

$$\text{optimality gap}(\%) = \frac{\text{agent cost} - \text{MILP cost}}{\text{MILP cost}} \times 100$$

A smaller optimality gap indicates a schedule closer to the mathematical optimum. The

Table 3.1 DSR phase mapping.

Phase	DSR step	Activity in this thesis	Output
1	Problem identification	SLR confirming no multi-agent ReAct HEMS addresses multi-appliance coordination across multiple price-volatility regimes	Gap statement; confirmed RQs
2	Define objectives	Specify measurable success criteria: optimality gap target, conflict resolution rate, latency, token cost	Objective specification
3	Design and development	Implement Appliance Profiler, Price-Aware Scheduler, Conflict Resolver, Orchestrator, and single-agent baseline in LangGraph	Working pipeline artifact
4	Demonstration	Run pipeline on five representative scenarios; confirm valid cost-reducing schedules	Demonstration results
5	Evaluation	Run 48 scenarios; apply Wilcoxon test; break down by scenario type and appliance type	Quantitative evaluation results
6	Communication	Write findings, design guidelines, limitations, and future work	Complete thesis

Wilcoxon signed-rank test is used to assess statistical significance of the difference between multi-agent and single-agent optimality gaps across all 48 paired scenarios. The test is appropriate because it is non-parametric, requires no assumption of normality, and is designed for paired comparisons. Statistical significance is assessed at $p < 0.05$ two-tailed. Practical significance is reported using effect size $r = Z/\sqrt{N}$.

4. System Design and Implementation

This chapter describes the design and implementation of the hierarchical ReAct multi-agent pipeline. It presents the problem definition and design objectives, the architecture of the complete pipeline, the specifications of each agent, the single-agent baseline implementation, the MILP ground truth solver, and the technology stack used throughout.

4.1 Problem definition and objectives

Based on the literature synthesis in Section 2.7 and DSR Phase 2, the artifact must satisfy five design objectives. First, multi-appliance coordination: the pipeline must correctly schedule two or more flexible appliances simultaneously without producing overlapping high-load windows. Second, price-volatility robustness: the pipeline must produce cost-reducing schedules across all six scenario types. Third, constraint compliance: every schedule must respect user-defined completion deadlines and household load limits. Fourth, interpretability: the full reasoning trace of every scheduling decision must be captured through LangSmith. Fifth, computational efficiency: the pipeline must complete scheduling within 60 seconds and under $0.10 API cost per scenario using GPT-4o-mini.

4.2 Architecture overview

The artifact is a LangGraph-based hierarchical pipeline organized as a directed state graph with four nodes, each implementing a ReAct agent or coordination logic. Outputs from each node are written to a shared LangGraph state object that persists across the full pipeline execution. The pipeline executes in the following sequence: the Orchestrator receives the user's scheduling request and initializes the shared state; the Appliance Profiler classifies flexible appliances; the Price-Aware Scheduler proposes cost-optimal start times; the Conflict Resolver eliminates overlapping high-load windows; and the Orchestrator synthesizes the final schedule for the user.

Figure 4.1 presents the complete pipeline architecture. Solid arrows indicate primary data flow; dashed arrows indicate state updates to the shared LangGraph state object. The single-agent baseline (dashed box, top right) receives identical inputs and is evaluated in parallel for comparison.

4.3 Agent specifications

The **Orchestrator** receives the household scheduling request, coordinates the three specialist agents in sequence, and synthesizes the final output. The Orchestrator does not

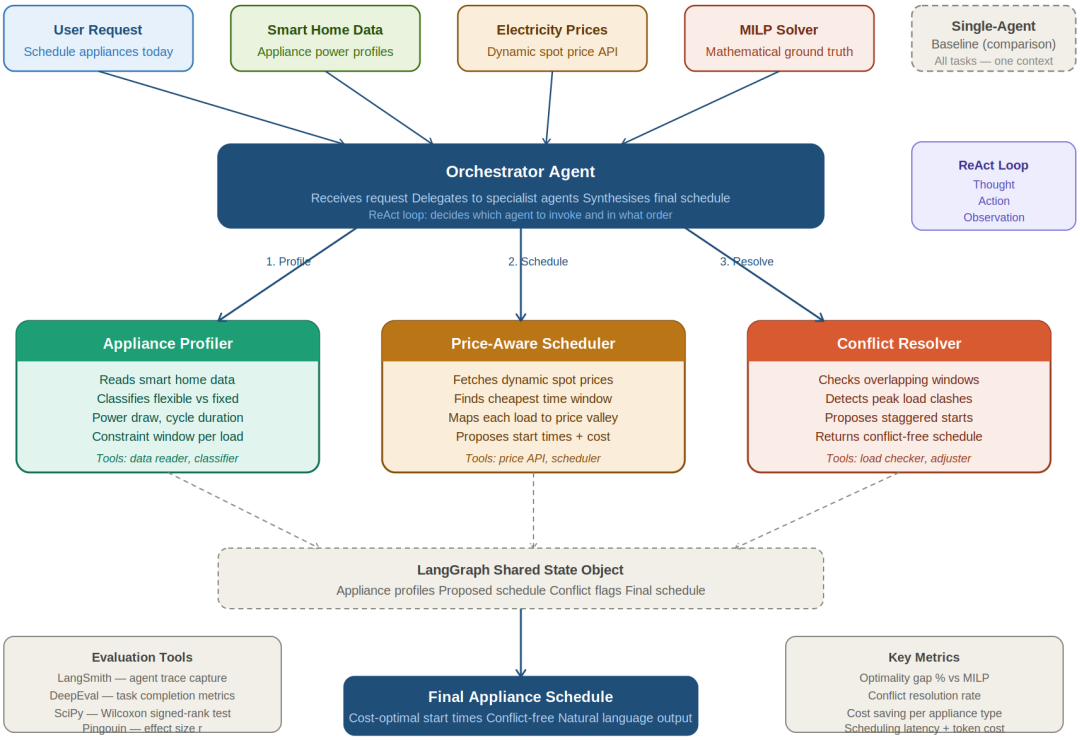


Figure 4.1. Hierarchical ReAct Multi-Agent Pipeline Architecture.

perform scheduling reasoning itself, keeping its context window small and decisions reliable across all scenario types.

The **Appliance Profiler** reads the household consumption data and classifies each appliance by shiftability. The classification distinguishes flexible loads — those with long cycle durations and user-defined completion windows such as washing machines and dishwashers — from fixed loads such as refrigerators and microwaves. The Profiler outputs a structured appliance profile containing power draw in Watts, typical cycle duration in minutes, and the constraint window for each flexible load.

The **Price-Aware Scheduler** receives the appliance profile and the day-ahead price schedule. It reasons through the price series to identify the cheapest available window for each flexible appliance within its constraint envelope and proposes a daily schedule with per-appliance start times and a total cost estimate.

The **Conflict Resolver** receives the proposed schedule and checks for overlapping windows where two or more appliances with high power draw would run simultaneously, exceeding the household's peak load threshold. If conflicts exist, the Resolver proposes staggered start time adjustments that eliminate the conflict while minimizing the increase in total schedule cost.

4.4 Single-agent baseline

The single-agent baseline is implemented using LangGraph's `create_react_agent` function with a single LLM instance and access to all tools available to the three specialist agents combined: the smart home data reader, appliance classifier, spot price tool, schedule calculator, cheapest window finder, load summation tool, conflict detector, and schedule adjuster. The baseline receives the same input prompt as the multi-agent pipeline. This design ensures a fair comparison: the only controlled difference between baseline and artifact is the presence or absence of role specialization.

The baseline is validated on five pilot scenarios before the full evaluation. Validation confirms that the baseline produces plausible but suboptimal schedules on simple scenarios and exhibits the coordination failures documented by Jonah et al. (2026) on complex multi-appliance scenarios, providing empirical motivation for the multi-agent design.

4.5 MILP ground truth solver

The MILP solver is implemented using PuLP and computes the mathematically optimal schedule for each scenario. The solver minimizes total daily electricity cost subject to three constraint types: each flexible appliance must complete exactly one cycle within its constraint window; no appliance cycle may be interrupted once started; and total household load at any time step must not exceed the household peak threshold. The solver is validated on the same five pilot scenarios as the baseline before the full evaluation begins.

4.6 Technology stack

Table 4.1 summarizes the technology stack used in this thesis.

Table 4.1 Technology stack.

Component	Tool	Cost
Orchestration framework	LangGraph v1.0+ (stable, Oct 2025)	Free, open source
LLM — commercial	GPT-4o-mini via OpenAI API	Under $3 for all 48 scenarios
LLM — open-weight	gpt-oss-20b via Ollama, local laptop GPU/CPU	Free — no API cost
MILP solver	PuLP with CBC (Python)	Free, open source
Agent tracing	LangSmith free tier	Free — native LangGraph integration
Statistical analysis	SciPy (Python)	Free — Wilcoxon test and effect size r

5. Evaluation and Results

This chapter presents the evaluation of the artifact, corresponding to DSR Phases 4 and 5. Section 5.1 describes the benchmark: the appliance and price data and the construction of the 48 evaluation scenarios across six price-volatility types. Section 5.2 uses the MILP solver alone to examine Finland's move to fifteen-minute electricity prices, quantifying how much of the achievable scheduling saving depends on fifteen-minute rather than hourly price information; this establishes where differences between schedulers can be expected before any agent is evaluated. Section 5.3 reports the demonstration of the pipeline on representative scenarios (DSR Phase 4). Sections 5.4 and 5.5 compare the optimality gaps of the multi-agent pipeline and the single-agent baseline across all scenarios and test the difference with the Wilcoxon signed-rank test. Section 5.6 breaks the cost savings down by appliance type. The agent results in Sections 5.3–5.6 are pending the full evaluation runs.

5.1 Dataset and scenario design

The evaluation benchmark combines real appliance consumption profiles from UK households with real Finnish day-ahead electricity prices. Each of the 48 scenarios describes one day in 96 fifteen-minute slots (00:00–24:00 local time) and consists of a price curve, a household base load, a set of flexible appliance cycles with user time windows, and a household peak load limit. The benchmark is generated by a deterministic, seeded build procedure, so it can be reproduced exactly from the source data.

5.1.1 Appliance data

Appliance profiles are drawn from the cleaned REFIT Electrical Load Measurements dataset (Murray et al., 2017; Murray & Stankovic, 2016), which records the whole-house aggregate and nine individually monitored appliances at 8-second intervals in 20 UK households between 2013 and 2015. The cleaned release corrects timestamps for daylight saving time and re-aligns appliance monitor columns where a monitor was moved between appliances during the study. The appliance-to-column mapping follows the dataset documentation.

Eighteen households are used. House 12 is excluded because it has no monitored washing machine, dishwasher or tumble dryer. House 19 is excluded after inspection: its washing-machine monitor recorded no heating phase in any of its 246 detected cycles (no fifteen-minute mean above 1.2 kW, against 22–70 % of cycles in the other households), so its profiles do not represent the water-heating load that dominates the cost and peak demand of a wash. Washer-dryers (houses 1, 8, 9 and 18) are not used as profiles, because they run washing and drying as a single cycle; their consumption is nevertheless removed from the

base load.

Individual appliance cycles are extracted from each monitor by thresholding the one-minute mean power at 20 W and merging pauses shorter than 30 minutes for dishwashers (whose programmes contain a passive drying pause) and 15 minutes for washing machines and tumble dryers (which are often reloaded within minutes). Cycles shorter than 20 minutes, longer than 300 minutes or below 0.1 kWh are discarded, and each remaining cycle is converted into a fifteen-minute mean power profile. This yields 6,172 washing-machine cycles from 19 monitored machines, 4,702 dishwasher cycles from 15 machines and 1,940 tumble-dryer cycles from 7 machines. To prevent heavily used appliances from dominating the benchmark, a scenario first draws a monitored appliance of the required type and then one of its cycles. The household base load is the median, per fifteen-minute slot of the day, of the aggregate minus all shiftable appliance monitors, computed after discarding readings that the dataset flags as inconsistent (sub-metered sum exceeding the aggregate).

5.1.2 Electricity price data

Day-ahead spot prices for the Nord Pool FI bidding zone were obtained for 1 January 2022 to 24 September 2026 from the public Sähkötin price service, which republishes Nord Pool data (Pakastin Oy, 2026). The values are in EUR/MWh excluding VAT and were verified against an independent source for the December 2022 crisis period. The service provides hourly values, which are repeated across the four fifteen-minute slots of each hour; days with a daylight-saving transition are dropped, leaving 1,718 complete days.

Scheduling cost is computed with the consumer price rather than the raw spot price. For spot price s, the consumer price per kWh is

$$p = (s + m + d + \tau)(1 + v),$$

with a retail margin $m = 0.5$ c/kWh, a distribution fee $d = 5.0$ c/kWh, the electricity tax $\tau = 2.253$ c/kWh and VAT $v = 25.5\,\%$. These are representative values for a Finnish household on a spot-price contract; a single VAT rate is applied to all days so that scenarios from different periods remain comparable. Using the consumer price keeps costs positive on days with negative spot prices, which keeps the optimality gap defined in Section 3.2 meaningful. Only the energy of the flexible appliances is costed, since the base load costs the same under every schedule.

5.1.3 Scenario construction

Table 5.1 summarises the six scenario types. Days are assigned to types by the within-day standard deviation of the spot price. Types A and B use disjoint days from the lowest quartile among days without negative prices, type C uses days between the 40th and 60th percentiles, and type D uses days in the top 15 % that fall in December–February. Type E uses days from December 2022, sampled from the 16 days with the highest maximum price. Type F uses days with at least two hours of negative prices. Eight days are sampled

per type.

Types C–F need up to five loads, while REFIT provides three flexible appliance types; a scenario may therefore contain two cycles of the same type, such as two washing loads. Each appliance has a user window (earliest start and deadline), and its naive start time is the earliest start. Types A–C use windows opening between 06:00 and 12:00 and lasting 8–14 hours. In type D all five appliances share one window from 08:00–10:00 until 22:00–24:00, so their cheapest start times coincide. In type E each appliance becomes ready 0–2 hours before the day's highest daytime price and has 4–8 hours of slack beyond its cycle, so the naive schedule runs into the price spike while avoiding it remains possible. This reflects that price spikes coincide with evening demand peaks, when households tend to start appliances. Type F windows open between 00:00 and 04:00 and last until midnight.

The household peak load limit, including the base load, is 3.5 kW. It represents a household power cap, for example one induced by a power-based distribution tariff, rather than the main fuse (typically 3×25 A, about 17 kW), which would never bind for these appliances. The build procedure verifies with the MILP solver that every scenario has a feasible schedule; this held for all 48 scenarios without relaxing the limit. Table 5.2 reports the resulting characteristics per type. A scenario requires coordination when running every appliance at its individually cheapest start time exceeds the peak limit; these 28 scenarios form the basis of the conflict resolution rate.

Table 5.1 Evaluation scenario design.

Type	Price regime	Appliances	Inst.	Research purpose
A	Stable — low volatility	2 appliances: dishwasher + washing machine	8	Baseline check — both systems should perform well
B	Stable — low volatility	3 appliances: dishwasher + washing machine + tumble dryer	8	Tests three-way conflict resolution
C	Moderate volatility	3 to 5 appliances mixed	8	Tests price-signal responsiveness
D	High volatility — winter peak	5 appliances, overlapping constraints	8	Most financially consequential condition
E	Price spike — Dec 2022 crisis	3 appliances, ready 0–2 h before the daily price peak, 4–8 h slack	8	Extreme Finnish price event robustness
F	Negative spot price — surplus	4 appliances, flexible windows	8	Tests exploitation of zero-cost windows

Table 5.2 Benchmark characteristics per scenario type.

Type	Price days	Mean appliances	Require coordination	Naive schedule within limit	MILP saving vs. naive
A	2022–2026	2.0	1/8	8/8	10.2 %
B	2023–2026	3.0	4/8	7/8	4.3 %
C	2023–2026	3.4	5/8	6/8	19.9 %
D	2022–2026 (Dec–Feb)	5.0	8/8	0/8	37.0 %
E	1–14 Dec 2022	3.0	4/8	3/8	14.4 %
F	2023–2025	4.0	6/8	5/8	7.8 %
All		3.4	28/48	29/48	

5.2 Value of 15-minute price information

Since 1 October 2025 the Nord Pool day-ahead market settles Finnish electricity in fifteen-minute periods. The main benchmark uses hourly prices, because its type E days date from the 2022 energy crisis. To examine the new market design, a second set of 48 scenarios (Q-A01 to Q-F08) was built with the same procedure from native fifteen-minute FI prices for 1 October 2025 to 25 September 2026 (358 complete days), obtained from the public price service of the Estonian transmission system operator, which republishes Nord Pool data (Elering AS, 2026). Its hourly averages match the hourly series used in Section 5.1. Because this period contains no crisis month, its type E days are drawn from the 16 days with the highest peak price (up to 65 c/kWh), and no day is used by more than one type. Prices genuinely vary within the hour: in 98 % of hours the four quarter-hour prices differ, with a median spread of 0.75 c/kWh.

For each Q scenario, two optimal schedules were computed with the MILP solver. The first is planned on the real fifteen-minute prices. The second is planned on hourly averages, which is all the price information a household had before the switch, and is then settled at the real fifteen-minute prices. Hourly prices make the four quarters of an hour tie, so ties are broken towards the earliest start. The cost difference between the two schedules is the saving that is only reachable with fifteen-minute price information. Table 5.3 reports the savings of both schedules relative to the naive schedule, the additional cost reduction from fifteen-minute information, and the share of the achievable saving that requires it; the “All” row weights scenarios by their cost.

Fifteen-minute price information lowers the optimal cost in 39 of the 48 scenarios, but the effect is modest: 1.2 % overall, or 3.7 % of the achievable saving. The remaining 96 % of the saving is available to a scheduler that plans on hourly prices, since most of the value lies in moving appliance cycles between hours rather than between quarters of an hour. The additional reduction grows with the intra-hour price spread (Spearman $\rho = 0.66$ across scenarios) and is largest on volatile days: 1.4 % in type D, 2.1 % in type E, and 10.2 % in the single largest case (Q-E03, 17 October 2025). Type C shows the largest relative share (16.4 %), because its hourly price differences are small, so intra-hour variation forms a larger part of the opportunity. In absolute terms the gain is about one cent per household

Table 5.3 Optimal schedule cost with hourly vs. 15-minute price information (MILP, Q scenarios).

Type	Intra-hour spread (c/kWh)	Saving, hourly plan	Saving, 15-min plan	Additional cost reduction	Share of saving needing 15-min	Scenarios improved
A	0.41	4.1 %	4.2 %	0.1 %	2.4 %	5/8
B	0.33	3.2 %	3.4 %	0.3 %	6.4 %	8/8
C	1.37	7.1 %	8.6 %	1.7 %	16.4 %	8/8
D	4.96	27.6 %	28.6 %	1.4 %	3.5 %	8/8
E	4.91	35.4 %	37.0 %	2.1 %	2.8 %	6/8
F	0.21	4.6 %	4.7 %	0.1 %	1.7 %	4/8
All	2.03	23.7 %	24.7 %	1.2 %	3.7 %	39/48

per day for the two to five cycles scheduled, and about three cents on type D days. The schedules nevertheless differ: 110 of the 163 appliance start times change between the two plans. These results indicate that the hourly main benchmark captures most of the scheduling problem, while any difference between the agents under fifteen-minute prices is expected to appear mainly in types C, D and E.

5.3 Demonstration results

Work in progress.

5.4 Quantitative results

Work in progress.

5.5 Statistical analysis

Work in progress.

5.6 Per-appliance breakdown

Work in progress.

6. Discussion

Work in progress. This chapter will interpret the results in relation to the three research questions, connect findings to the agentic AI trends reviewed in Chapter 2, and address the limitations and directions for future research.

6.1 Interpretation of research questions

Work in progress.

6.2 Limitations

Several limitations of this study warrant acknowledgement. First, the evaluation is entirely simulation-based using historical data. While the use of real consumption profiles and real Nord Pool FI prices provides stronger ecological validity than synthetic scenarios, the study does not evaluate the pipeline in a live deployment where appliance states change dynamically and user preferences may deviate from scripted constraint windows. Second, the REFIT dataset was collected from UK homes between 2013 and 2015 and may not fully reflect the appliance configurations and usage patterns of contemporary Finnish households. Third, LLM outputs are non-deterministic: two runs of the same scenario may produce different schedules. To mitigate this, each scenario is run three times and results are averaged, but residual variance from non-determinism remains. Fourth, the study evaluates only three appliance types; generalizing findings to electric vehicle chargers, heat pumps, or battery storage requires additional evaluation.

6.3 Future research

Several directions for future research emerge from this study. First, extending the pipeline to include thermal loads — electric space heating, heat pumps, and domestic hot water systems — addresses the gap explicitly identified by El Makroum et al. (2026). Second, implementing the price API and data reader as MCP servers would enable reuse by future researchers without custom integration code, contributing to the emerging standardization of agent tool interfaces. Third, a live deployment study involving Finnish households on spot-price contracts would provide ecological validity that simulation-based evaluation cannot. Fourth, extending the multi-LLM comparison to include Qwen-3-32B and Phi-4 would provide a more complete picture of the SLM landscape for residential scheduling tasks.

7. Conclusions

Work in progress. This chapter will summarize the main contributions of the thesis, explicitly answer RQ, SQ1, and SQ2 using evidence from Chapter 5, and state the implications for academic research on multi-agent LLM systems and practical deployment of intelligent demand flexibility tools in Finnish residential settings.

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